

Short Description of the Scene

This assignment implements tangent-space normal mapping in OpenGL using custom GLSL vertex and fragment shaders to demonstrate how surface detail can be simulated without increasing geometric complexity. The scene renders two identical rotating sphere models side-by-side under the same dynamic lighting conditions: the left sphere uses standard interpolated vertex normals, while the right sphere applies a tangent-space normal map. The vertex shader constructs a proper TBN (Tangent, Bitangent, Normal) matrix to transform sampled normal vectors from texture space into world space, and the fragment shader performs per-fragment Blinn-Phong lighting using both a diffuse texture and a normal map. This side-by-side configuration clearly highlights the visual difference between flat shading and normal mapping while keeping geometry, materials, and lighting identical. Real-time controls implemented with ImGui allow adjustment of light position, rotation speed, ambient intensity, and normal strength to dynamically observe the lighting response. External libraries used include GLFW for window and context management, GLAD for OpenGL function loading, GLM for linear algebra, Assimp for importing the glTF model, stb_image for texture loading, and Dear ImGui for interactive parameter control.